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# EBU5608 Product Development and Management

Topic 2 – Digital Transformation and Digital Products

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## Agenda

- Digital Transformation
- Digital Product
- Digital Project Development
- Digital Project Management

- 60 minutes lecture
- 15-20 minutes in-class activity
- 5-10 minutes for Q&A

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## Digitisation vs. Digitalisation vs. Digital Transformation

- **Digitisation** is the process of converting information from analogue to digital.
- **Digitalisation** is the process of using digitised information to make established ways of working simpler and more efficient.
- **Digital transformation** is the process of using digital technologies to create new — or modify existing — business processes, culture, and customer experiences to meet changing business and market requirements

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## Digital Transformation

**Digital transformation** is:

" *a process of reinvention. It looks at leveraging digital technologies and processes to build new systems, cultivate a digital mindset and exploit new opportunities*" [1]

" the *ability of the organisations to build and deliver products and services through software*" [2]

" *It begins and ends with how you think about, and engage with, customers*" [1]

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## Types of Digital Transformation [3]

There are four types of digital transformation these include:

- **Process Transformation** – aims to process such as data, analytics, AI, and any process that can work towards lowering costs and driving operational efficiency in the business.
- **Business Model Transformation** – aims to make fundamental changes in how a business or organization runs which can include personnel, processes, and technology.

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## Types of Digital Transformation

- **Domain Transformation** - This area offers a great opportunity to move into a new domain or area that a business may not have explored before by acquiring new technologies.
- **Cultural/Organizational Transformation** - This is about redefining mindsets, processes, capabilities and skills for a digital world. It's about driving digital transformation forward through growth initiatives that are grounded in a new culture and way of thinking.

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# Digital Transformation – Examples

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- Many traditional products and services have already been completely disrupted and transformed by digital technologies, such as
  - **Finance** – Fintech platforms like Revolut and Stripe have revolutionized banking and payments, offering seamless digital transactions, mobile-first banking, and AI-driven financial insights.
  - **Automotive** – Tesla has disrupted the car industry with over-the-air software updates, autonomous driving advancements, and direct-to-consumer sales, bypassing traditional dealerships.
  - **Education** – E-learning platforms like Coursera and Duolingo have made education more accessible, using AI to personalize learning experiences and provide real-time feedback.

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# Digital Transformation – Examples

| Traditional Marketing Channel | Digital Marketing Channel | Transformational Impact | Recent Examples |
|---|---|---|---|
| **Print materials** | Digital materials (eBooks, PDFs, interactive content) | Reduces cost of print and distribution; enables real-time updates and interactive engagement | Digital whitepapers, interactive reports from companies like HubSpot and McKinsey |
| **Print mail campaigns** | Email campaigns & AI-powered automation | Reduces cost of print and postage; enables hyper-personalization and automation at scale | AI-driven email personalization by platforms like Mailchimp and Klaviyo |
| **Print/billboard advertising** | Social media & programmatic advertising | Enables real-time, data-driven targeting; lookalike audience targeting for greater efficiency | AI-powered ad targeting by Meta (Facebook & Instagram) and Google Ads |
| **Brick-and-mortar storefront** | Website/e-commerce site & AI-driven personalization | Eliminates rent/utilities; provides accessibility and scale; AI-driven recommendations boost engagement | Shopify, Amazon AI-powered product recommendations, TikTok Shop |
| **Loyalty Club Card** | Mobile App & AI-driven customer engagement | Reduces signup friction and card printing costs; enables real-time, personalized promotions and push notifications | Starbucks Rewards App, McDonald's mobile rewards, AI-powered recommendations by Sephora and Nike apps |

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# Digital Transformation – Across Industries [3]

| Sector | Key Areas of Digital Transformation | Challenges & Strategies |
|---|---|---|
| **Governance** | E-government and digital services<br>Digital identities and blockchain | 70% of organizations lag behind the private sector in digital capabilities |
| **Healthcare** | Remote patient monitoring<br>AI-based automation (e.g., image-guided therapy)<br>Tele-ICUs and tele prescribing | Adapting to digital health technologies |
| **Financial Services** | Touchless transactions<br>Hyper-personalization<br>Super apps | Accelerated digital change in response to market trends |
| **Manufacturing** | Robotics<br>AR for assembly lines<br>3D printing<br>B2B e-commerce<br>IoT for real-time data | Accelerated by the pandemic |
| **Small Business** | Improving operations, customer experience, market expansion, sales, marketing, talent sourcing<br>Automating processes, investing in talent & digital skills | 72% accelerating digitalization<br>Specialized training programs |
| **HR** | Adoption of HR software<br>New ways of working (remote, hybrid)<br>Digital learning platforms<br>People analytics<br>Cloud-based HR | Expected to reach $43 billion by 2028<br>Optimize workplace experience, efficiencies, upskill & train staff, workflow automation, talent management |

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# Digital Transformation – Trends [4]

- **5G** – Unlocks ultra-fast, low-latency connectivity, enabling advanced services such as real-time IoT applications, immersive AR/VR experiences, and autonomous systems.
- **Zero-Trust Security** – Strengthens cybersecurity by continuously verifying users, devices, and applications across networks, reducing the risk of breaches and insider threats.
- **Hyperautomation** – Automates complex business and IT processes using AI, Robotic Process Automation (RPA), and intelligent workflows to increase efficiency and reduce costs.
- **Predictive Analytics** – Leverages AI and Machine Learning (ML) to analyze historical and real-time data, allowing businesses to anticipate trends, optimize operations, and improve decision-making.
- **Generative AI** – Creates new content—including text, images, audio, and video—by learning from existing datasets, revolutionizing content creation, marketing, and software development.

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## Guidelines for a successful Digital Transformation

- Understand your technology
- Embrace Cultural Change
- Consider a new digital business model
- Digital upskilling
- Ensure Collaboration
- Top Management Support

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## Digital Product

**Digital products**:

- **no physical** form, exist only in the digital realm,
- **intangible** items delivered **electronically**,
- anything that can be **downloaded** and **used digitally** can be considered a digital product,
- sold online or through brick-and-mortar retailers,
- can be easily updated or modified to keep up with changing technology and trends because they're intangible,
- often come with a license that allows a customer to use them in unlimited ways.

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## Will everything become digital?

In practice, most products and experiences are **hybrid**.

- Replace a physical product with digital copy,
  - e.g. paying with coins becomes paying through an app,
  - a paper entrance ticket becomes a QR code on our phone.

The **added value** of simply a digital equivalent of a previously analogue product is often **rather small**.

A digital reproduction of a physical product is usually not the best way to go.

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## Top 10 digital products [5]

| Digital Product | Description | Platform/Market |
|---|---|---|
| **AI-Generated Art and Designs** | Unique, customizable digital art and design templates. | Marketplaces, Etsy, Self-Publishing |
| **Online Courses** | Courses in coding, marketing, personal development, and more. | Udemy, Thinkific |
| **Templates and Digital Assets** | Ready-to-use templates for social media, budgeting, project management. | Canva, Etsy, Creative Market |
| **Printables** | Digital planners, trackers, and study aids. | Etsy, Gumroad |
| **Music and Audio Tracks** | Royalty-free music, sound effects, pre-recorded meditations. | AudioJungle, SoundCloud |
| **AI Tools and Prompts** | Tools and prompt libraries for automating tasks. | Tech communities, online platforms |
| **Digital Art and NFTs** | Blockchain-based or traditional digital art, illustrations, and prints. | OpenSea, Etsy, Self-publishing |
| **Online Learning Modules** | Structured courses, assignments, notes, and certificates. | Coursera, Udemy, Thinkific |
| **Specialized Financial Planning Tools** | Spreadsheets and software templates for budgeting, tax, and investment tracking. | Google Sheets, Excel, Templates |
| **Sustainable Living Guides** | Digital guides for sustainable living practices, zero-waste, and gardening. | Etsy, Self-publishing |

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## Why digital products?

- Customer happiness is how you win in business. Modern customer expectations are being driven by largely digital technology and digital innovations.
- Low investment, (potentially) high returns
- More profitable than physical goods
- No inventory, shipping or rent hassle
- Automated delivery for passive income
- Serve a niche at scale
- Digital products offer unique ways to communicate directly with the customers.

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## Digital Project Development

Six phases of digital product development [6]:

- **Phase 1. Discovery**
- **Phase 2. Ideate**
- **Phase 3. Test**
- **Phase 4. Execute**
- **Phase 5. Launch**
- **Phase 6. Grow**

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## Phase 1. Discovery

***Discovery*** is the process of identifying the problem to be solved, making sure the problem is worth solving, and envisioning the solution to that problem.

- **Stay Alert in your Everyday Life** - anything that bothers you might be pointing you in the direction of a problem that no one has ever thought to solve.
- **Find a Way to Improve an Existing Product** - a product that already exists to solve the problem is a good indicator that someone thought it was worth solving.
- **Create a Unique Selling Proposition**. Could you create a cheaper version? A luxury version? An automated version?
- **Research Consumer trends** like eco-friendliness, social justice, plant-based diets, fitness, etc.

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## Phase 2. Ideate

***Ideate.*** The goal is to brainstorm possible solutions to the problem identified in the discovery phase, creating a strategy for how to build a product that will solve that problem.

- **Brainstorming**. Get together a cross-functional team and leverage group brain power by bouncing ideas off of each other. Keep the group diverse, but not too large.
- **Prototyping**. Physical products are often prototyped, but digital products can be prototyped too in the form of a non-functional model version that allows stakeholders to visualize the final product.

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## Phase 2. Ideate - continue

- **Storyboarding** is well-known in the movie industry, but by drawing the user's interaction with the product as a story with a beginning, a middle, and an end, you can tease out game-changing features to add.
- **Five Whys** analysis, popularized by Toyota, involves starting with the statement of the problem and then asking at least five sequential "why" questions in response to the statement, with each why springing from the previous "why." The goal is to get at the root essence of the problem.
- **Method 6-3-5**. Meant to get the creative juices flowing, 6-3-5 involves a team of six people writing down three ideas in five minutes, then passing the results in a circle around the table so the other five team members can add on to each idea.

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## Phase 3. Test

***Test.*** The testing phase is all about gathering data, refining and improving your idea, and gathering more data until you have a sharp idea. The testing phase should involve at least four steps, known as the *lean validation process* …

- **Validate the Problem.** Does the target market agree that the problem is worth solving?
- **Validate the Market.** Is the market for the product robust enough to justify the development project?
- **Validate the Product.** Does your product actually solve the problem?
- **Validate Willingness to Pay.** Is the target market willing to pay a price for a solution to this problem that would justify developing the product?

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## Phase 4. Execute

***Execute.*** This stage aims to develop the"***most valuable player***" namely the **Minimum Viable Product (MVP)**.

- **MVP** is the product with the bare minimum of features and functionality needed to solve the problem identified in the discovery phase and validated in the testing phase.
- In some ways, the MVP will be the ultimate testing vehicle, gathering feedback you can use to refine the product in the race to its final form.
- A strong MVP may even capture VC imaginations and attract investment.

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## Phase 5. Launch

Once your MVP is ready to go, it's time to ***launch***. A digital product launch usually means putting the MVP on the market and giving customers their first crack at your solution. Several approaches to this launch:

- **Soft Launch**—The MVP is made available to a limited segment of the target market, known as "early adopters."
- **Hard Launch**—The MVP is made available to the market at large with a full marketing push (not recommended for MVP launches).
- **Dark Launch**—The MVP has already been launched, but new features get a "dark" launch to a select percentage of users to test their behavior in reaction to the new feature.

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## Phase 6. Grow

As refine the digital product into its final form, based on the feedback from the MVP launch, it's time to consider the **growth** or **scaling phase**. One of the reasons to do an MVP before a full product launch is so that your product *has* room to grow and scale. The tasks relevant to this phase include:

- **Collecting and Analyzing Feedback.** The customers' interaction with the MVP will provide valuable data to use in improving the digital product.
- **Preparing to Scale.** This is basically preparing for success. If you had 100,000 orders overnight, could you handle that? Shore up automations, SOPs, scalable cloud services, and support staff contingencies in the event that need to go big, fast.

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## Phase 6. Grow

- **Zeroing in on the Optimal Price.** The MVP data could suggest that got the price a little too high, or even too low, especially with added features.
- **Scaling Up Marketing.** the MVP leads can serve as valuable customer personas to create lookalike audiences to market to.
- **Testing and More Testing.** Phase 3 really never stops. At every stage of growth, keep validating the old hypotheses and new ones to make the product as viable as possible.

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## Digital Project Manager [7]

- A digital product requires a mix of skills in addition to product management.
- A **digital product manager** is the glue that brings together many facets of a successful digital product—customers, design, engineering, operations, sales, marketing, finance, compliance, legal, and more.
- An analytical mindset and a combination of skills like
  - **communication**,
  - **patience**,
  - **decision-making**,
  - **curiosity**,

  help build the right perspective to being a digital product manager.

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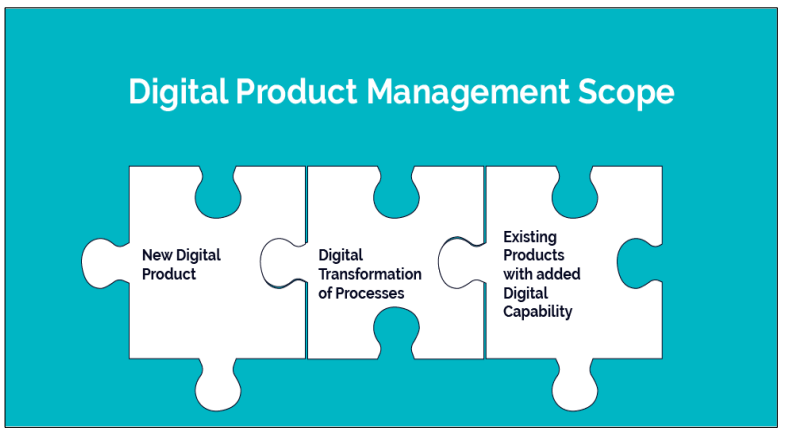
## Digital Project Management [7]

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## Project Team for Digital Products

- A project team for digital product is made up of
  - **Developers** or **engineers** – the people who will code, test and deploy the digital application that will be used by customers.
  - **Experts** in **customer or user experience** who focus on how the product and associated services will be used by the customer, and who create the user interface, services and other interactions with the customers
  - **Sales and marketing experts**, who will actually get customers to use the product.

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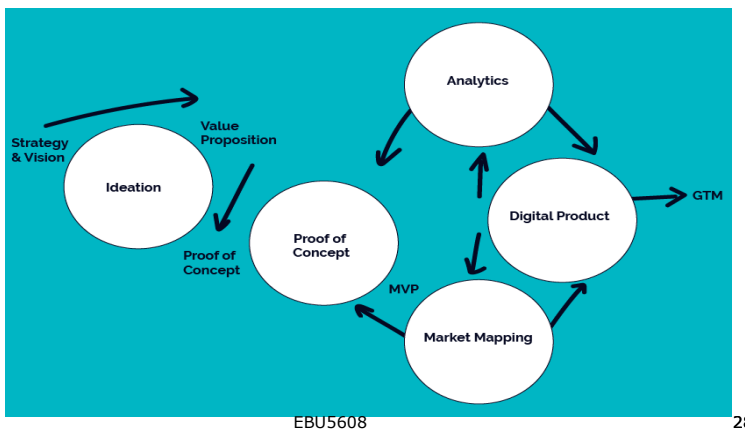
## Digital Project Management Lifecycle [7]

Digital product managers' responsibilities are spread throughout the product life cycle.

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## Evaluate the opportunities for digital transformation

- Are there any opportunities to do things more simply thought digital?
- What is our technology really capable of, and how can we adapt our business and processes to make the most of our technology investments?
- What's good about the physical experiences that you do not want to lose?
- What services can we provide around the physical and/or digital product to enhance the overall customer experience and value?

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## Summary of mistakes to avoid [3]

- Having the wrong data
- Resistance from staff
- Underestimate costs
- A lack of commitment
- A lack of skills

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# References

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